PREPRINT · POPULATION GENETICS

The Jesus DNA Hypothesis: A Quantitative Reassessment of Patrilineal Survival and Carrier Numbers for a 1st-Century Judean Founder

Alexi Choueiri
Independent researcher
alexichoueiri@gmail.com

Version 1.0 · 11 November 2026
Licensed under CC BY 4.0

ABSTRACT

Background. If a man living in 1st-century Galilee or Judea had a son, his Y chromosome could in principle still be carried by men alive today. A natural first estimate, scaled from famous lineages such as that attributed to Genghis Khan, puts such carriers in the hundreds of thousands, concentrated in the Levant. This paper tests that estimate, using the patriline of Joseph of Nazareth as a hypothetical example (the "Jesus DNA hypothesis").

Methods. The survival and size of a single founder's male line were modelled as a Galton–Watson branching process, with 2 million simulated founders per scenario across 65 or 80 generations, 1- to 30-fold population growth and two levels of variance in sons per man. Expected Y-chromosome and mitochondrial mutations were computed from pedigree-based rates.

Results. Only 2–10% of male lines survive 65–80 generations. A surviving line reaches a median of 24–205 living men, and none of 10 million simulated founders exceeded 4,000. A line of 200,000 men would require reproductive advantage on the scale of empire-driven lineages. A surviving Y chromosome would still match the founder's at about 99.9998% of sites (~109 mutations). For any given man in the Levant today, the chance of carrying such a line is well under 1 in 100,000.

Conclusions. The line most likely died out. If it survived, it is most likely carried by a few hundred living men, probably clustered in a few families, not by hundreds of thousands. Model code is released for reproducibility.

Keywords · Y chromosome · mitochondrial DNA · uniparental markers · lineage extinction · Galton–Watson branching process · genetic genealogy · Levant · science communication

1 Introduction

Most of a person's DNA is reshuffled every generation. Two pieces are not. The Y chromosome passes from father to son, and mitochondrial DNA (mtDNA) from mother to child, almost unchanged. Each works like a receipt: a record of one unbroken line of descent that can, in principle, last for thousands of years. These markers have been used to test real lineage claims, from the Jewish priesthood to the Mongol expansion (Zerjal et al. 2003).

© 2026 Alexi Choueiri. This work is licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0). Model code and output accompany this preprint in its Zenodo record.

This paper asks a simple question of that receipt. If one ordinary man in 1st-century Galilee or Judea had a son, how many men alive today would carry his Y chromosome? The example used throughout is the patriline of Joseph of Nazareth, carried forward through the brothers named in Mark 6:3 under traditions that treat them as Joseph's sons. Call it the **Jesus DNA hypothesis**: the idea that this line survives today in a substantial number of men.

A natural first estimate is large. Famous lineages are carried by millions, so a 2,000-year-old line might plausibly be carried by about 200,000 men across the Levant and its diaspora. This paper tests that estimate against a standard model of how male lines grow and die out.

Three distinctions frame the analysis:

1. **Genealogical versus genetic ancestry.** Anyone alive 2,000 years ago who left descendants is probably a genealogical ancestor of a very large share of people living today (Rohde, Olson & Chang 2004). Carrying that person's Y chromosome is far rarer, because it follows a single father-to-son path through the family tree.
2. **Haplogroup versus individual.** Sharing a haplogroup such as J1 shows deep common ancestry, usually thousands of years older than any named person. It does not identify an individual founder.
3. **Modelled versus observed descent.** No Y-chromosome data exist for any relevant 1st-century individual, so every carrier figure here is a model result, not an observation.

This paper makes no religious or theological claim. The founder is treated only as a hypothetical 1st-century man.

2 Methods

Lineage model. Survival of a single founder's male line was modelled as a Galton–Watson branching process, in which each man has an independent random number of sons. Survival probabilities were computed exactly by iterating the offspring generating function. Lineage sizes were estimated from 2,000,000 simulated founders per scenario (seed 20261111). Scenarios varied three inputs:

- **Generations:** 65 (a 31-year male generation interval over the 1,996 years from 30 AD to 2026) and 80 (a 25-year interval).
- **Population growth over the period:** none, 10-fold and 30-fold, applied as a constant per-generation mean number of sons.
- **Variance in sons per man:** Poisson (variance equal to mean) and overdispersed negative binomial (shape $k = 1$, variance about twice the mean).

Mutation arithmetic. Expected point mutations along one 1,996-year patriline used the pedigree-based Y rate of 8.71×10^{-10} per site per year (Helgason et al. 2015) and the complete Y length of about 62.5 Mb (Rhie et al. 2023). The mtDNA comparison used 1.665×10^{-8} per site per year over 16,569 bp (Soares et al. 2009).

Regional illustration. To show what line sizes mean on the ground, Section 5 applies an illustrative regional split of a surviving line's carriers across the Levant and its diaspora (Figure 5). The shares are assumptions chosen for illustration, not model results.

3 The Y chromosome as a record

A surviving Y chromosome would still be a near-perfect copy of the founder's after 2,000 years. Table 1 gives the figures.

Table 1. Uniparental markers over one 2,000-year line of descent.

Quantity	Value	Source or basis
Y chromosome length	~62.5 million base pairs (~1% of a man's genome)	Rhie et al. 2023
Male-specific (non-recombining) share	~95%; mostly non-coding, with dozens of genes including SRY	Rhie et al. 2023
Generations since 30 AD	~64–65 at 31 years per generation	Calculation
Y point mutations along the line	~109 across the whole chromosome	Helgason et al. 2015 rate
Y identity to the founder	~99.9998%	Calculation
mtDNA length	16,569 base pairs	Standard reference
mtDNA mutation rate per site	~19 times the Y rate	Soares et al. 2009
mtDNA mutations along the line	~0.55; about 58% of maternal lines unchanged	Calculation

Near-identity is not identification, however. Every male line is near-identical to its own founder, and two unrelated men from the same region typically differ at only hundreds to a few thousand Y positions. A living man's Y chromosome could be tied to a particular 1st-century founder only through a sample from that founder or a well-documented genealogy, and neither exists. The question is therefore not whether a carrier could be recognised, but how many carriers could exist.

4 Results

Most 2,000-year male lines die out, and those that survive stay small. Under neutral inheritance, a surviving line reaches tens to a few thousand living men, roughly 50 to 8,000 times fewer than 200,000.

4.1 Will the line survive?

About 37% of lines end in the first generation, when a man has no sons, and about 84% are gone within ten generations. By the present, only 1.5–7.5% survive, depending on population growth and on how unevenly sons are spread among men. If the founder is known to have had at least one son, the figure rises to about 3–12% (Figure 1). In every scenario, the most likely outcome is that the line no longer exists.

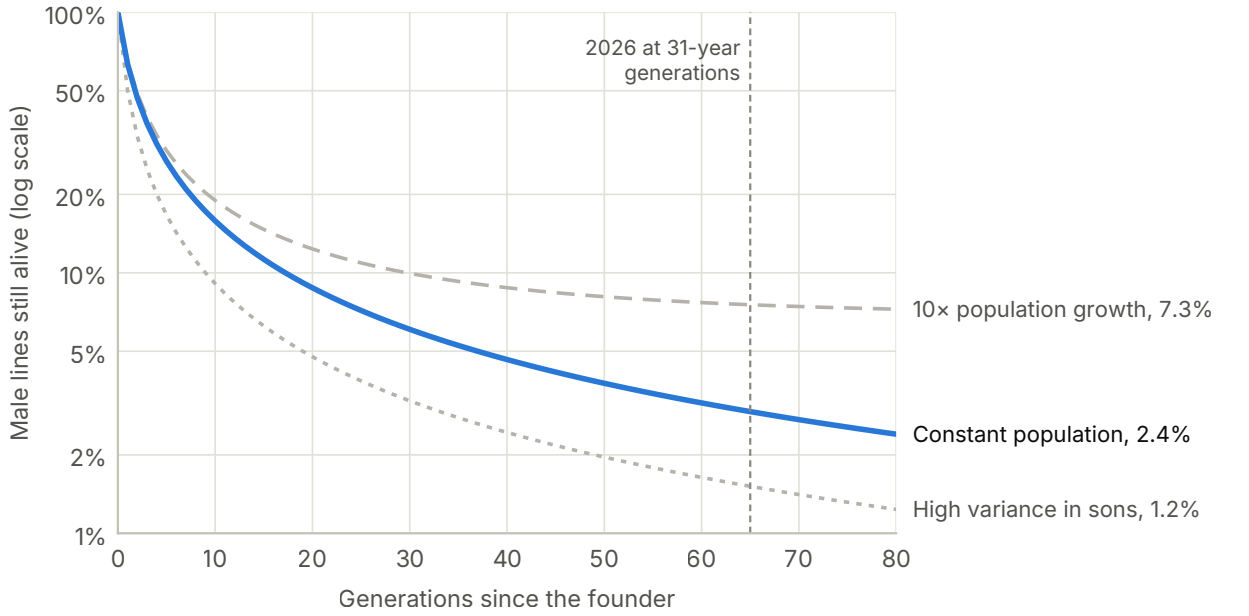


Figure 1. Most male lines die out within a few hundred years. Share of founders whose male line is still alive, by generation (log scale), from exact branching-process calculations. The per-generation growth rate is set for 65 generations and extended to 80. The dashed vertical line marks the present at a 31-year generation interval.

4.2 How large is a surviving line?

Intuitive estimates of a surviving line's size tend to be large. Three natural scenarios give 50,000–100,000 men if descendants stayed in Galilee villages, 200,000–500,000 if they spread across the Levant and its diaspora, and around a million with strong branching and later migration. Figure 2 sets them beside the model's result.

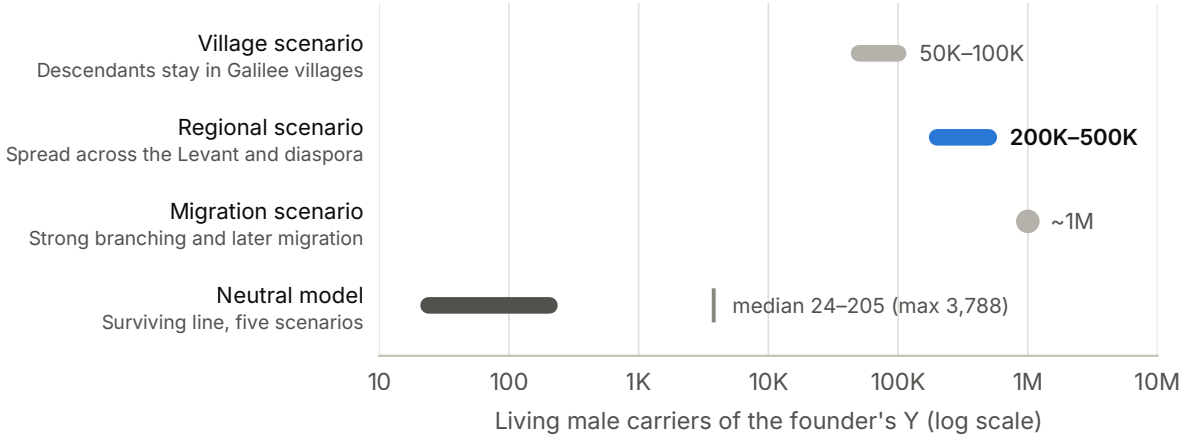


Figure 2. Even the smallest intuitive estimate lies far above a neutral line. Three intuitive scenarios for the number of living carriers, compared with the range of median surviving-line sizes across the model's five scenarios (tick: largest of 10 million simulated founders). Log scale.

Figure 3 shows the model's results in detail.

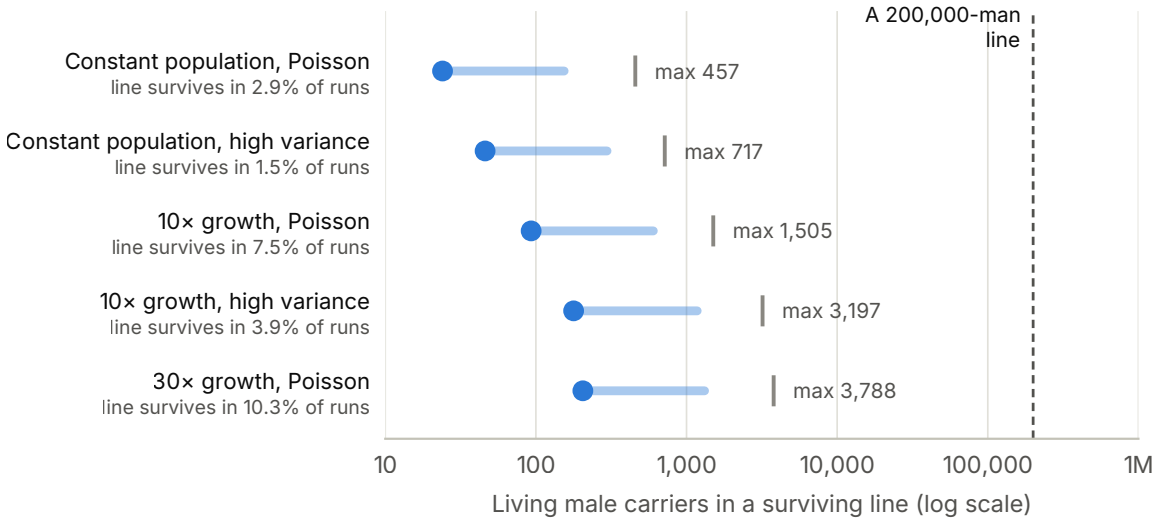


Figure 3. Surviving male lines reach hundreds of men, not 200,000. Living men carrying one founder's Y chromosome after 65 generations, among lines that survived. Dots mark the median, bars extend to the 99th percentile, and ticks mark the largest of 2,000,000 simulated founders per scenario (seed 20261111). The dashed line marks a 200,000-man line.

The reason for the gap is simple arithmetic. On average, a founder's male line grows only as fast as the population: in a stable population, one man's line averages one man today. Because most lines die out, the few survivors are larger than average, but only enough to make up for the others: about 34–290 men across the scenarios. None of 10 million simulated founders reached 4,000 carriers.

Put another way, the men of the Levant today are shared out among the tens of thousands of 1st-century male lines that survived. For one line to hold 200,000 men, it would have to take the share of about a thousand others.

4.3 What a large line requires

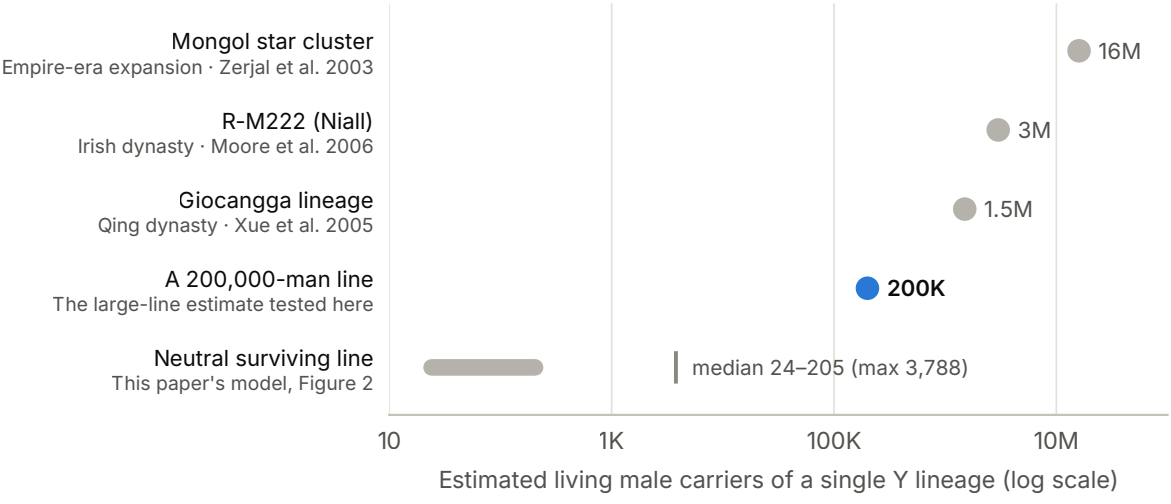


Figure 4. A 200,000-man line belongs with dynasty-driven lineages. Estimated living male carriers of single Y lineages (log scale). Published estimates: Zerjal et al. (2003), Moore et al. (2006), Xue et al. (2005). The neutral range is the span of median line sizes across the scenarios in Figure 3, with the tick marking the largest simulated line.

The lineages that do reach hundreds of thousands or millions of men share one feature: power. The Mongol "star cluster", the Qing founder Giocangga's line and the Irish R-M222 lineage all spread through empire or dynasty, which gave a few families many more sons than others for centuries. The

attributions to Genghis Khan and to Niall are themselves debated (Wei et al. 2018), but the mechanism is not. A carpenter's family in 1st-century Nazareth would have had no such advantage, so its line, if it survived, should look like an ordinary one.

5 Where carriers would live

A surviving line would most plausibly remain concentrated where the founder's descendants stayed. Figure 5 sets out an illustrative split across the Levant and its diaspora, with the communities where carriers would most plausibly live, and converts it into numbers of men.

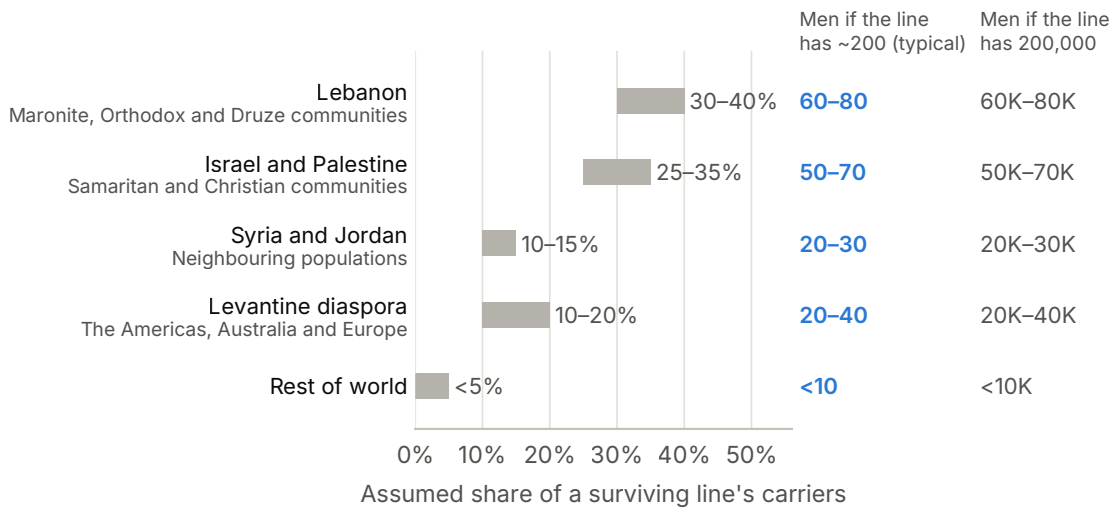


Figure 5. Where a surviving line's carriers might live, and how many men that means. Bars show an assumed share of carriers by region (an illustration, not a model result). The right-hand columns convert those shares into men for a typical surviving line of about 200 men (Figure 3) and for a 200,000-man line: a few dozen men per region in the first case, tens of thousands in the second.

For a typical surviving line, that means a few dozen men in each country, probably clustered in a handful of extended families and villages rather than spread evenly. Such a cluster could be locally noticeable while remaining extremely rare across the region. Once the chance that the line survived at all is included, the probability that any given man in the Levant carries it is well under 1 in 100,000. In a room of 100 men from the region, almost certainly none would.

6 Discussion

The receipt image holds: a surviving Y chromosome would still be about 99.9998% identical to the founder's after two thousand years. What fails is the scale. Male lines mostly vanish, and those that survive without social advantage stay small. The famous large lineages are large precisely because they had that advantage.

The same logic applies to the maternal line. If Mary had daughters, and they had daughters in an unbroken chain, their maternal-line descendants would carry her mtDNA, most of them with no changes at all. Female lines face the same extinction risk, however, so the same conclusion follows: most likely extinct, and small if not.

Finally, these results concern only two narrow receipts. The rest of the genome is inherited through every ancestor. In that ordinary family-tree sense, a very large share of people in the Levant today, and well beyond it, probably descend from 1st-century Galileans and Judeans, whatever happened to any single father-to-son line.

The answer to the opening question is therefore short. The line probably did not survive. If it did, it is most likely carried today by a few hundred men, each with a Y chromosome almost identical to the founder's, and not by hundreds of thousands.

7 Limitations, ethics and availability

Model limitations. The Galton–Watson model treats men as independent and the population as unstructured. Real populations add local density regulation, migration, endogamy and village-level drift, which can enlarge some lines locally; but none of these produces the several-thousand-fold excess needed for 200,000 carriers without differential reproductive success. Generation interval, growth factor and offspring variance were bracketed rather than estimated for the 1st-century Levant, and mutation counts use present-day pedigree rates. Section 5's regional shares are illustrative.

Ethical considerations. Claims of descent from religious figures carry deep personal, communal and religious significance. Numerical estimates of "carriers" can be read as genetic verdicts on particular families or communities. This paper makes no claim that any living person or group descends from any historical figure, and its figures should not be used to support or dispute such claims. No personal or genetic data were collected.

Data and code availability. The lineage model (`lineage_survival_model.py`, Python 3 with NumPy, fixed seed 20261111), its output (`lineage_survival_results.json`) and the figure script (`make_figures.py`) accompany this paper in its Zenodo record. They reproduce every model-derived number and figure.

Funding and competing interests. None declared.

8 Statement on the use of AI assistance

A generative AI assistant was used to support literature retrieval, drafting and editing of the text, writing of the simulation code and preparation of the figures.

References

Helgason, A., Einarsson, A. W., Guðmundsdóttir, V. B., et al. (2015). The Y-chromosome point mutation rate in humans. *Nature Genetics*, 47(5), 453–457. doi:10.1038/ng.3171

Moore, L. T., McEvoy, B., Cape, E., Simms, K., & Bradley, D. G. (2006). A Y-chromosome signature of hegemony in Gaelic Ireland. *American Journal of Human Genetics*, 78(2), 334–338. doi:10.1086/500055

Rhie, A., Nurk, S., Cechova, M., et al. (2023). The complete sequence of a human Y chromosome. *Nature*, 621, 344–354. doi:10.1038/s41586-023-06457-y

Rohde, D. L. T., Olson, S., & Chang, J. T. (2004). Modelling the recent common ancestry of all living humans. *Nature*, 431, 562–566. doi:10.1038/nature02842

Soares, P., Ermini, L., Thomson, N., et al. (2009). Correcting for purifying selection: an improved human mitochondrial molecular clock. *American Journal of Human Genetics*, 84(6), 740–759. doi:10.1016/j.ajhg.2009.05.001

Wei, L.-H., Yan, S., Lu, Y., et al. (2018). Whole-sequence analysis indicates that the Y chromosome C2*-Star Cluster traces back to ordinary Mongols, rather than Genghis Khan. *European Journal of Human Genetics*, 26(2), 230–237. doi:10.1038/s41431-017-0012-3

Xue, Y., Zerjal, T., Bao, W., et al. (2005). Recent spread of a Y-chromosome lineage in northern China and Mongolia. *American Journal of Human Genetics*, 77(6), 1112–1116. doi:10.1086/498583

Zerjal, T., Xue, Y., Bertorelle, G., et al. (2003). The genetic legacy of the Mongols. *American Journal of Human Genetics*, 72(3), 717–721. doi:10.1086/367774